MQGT–SCF Execution Report (2026)

Local Replication Runs, Null Test, and Protocol Upgrades

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March 2026

Abstract

This report documents executed replication commands, produced artifacts, and test outcomes after upgrading the QRNG channel from a placeholder contract pipeline to a statistical hypothesis-testing pipeline. The run set includes a canonical run, a null-fixture adversarial control, and deterministic rerun checks. We also record protocol-level upgrades (operational E-label estimator, permutation-test significance, OSF-ready preregistration pack, and reproducible TeX build notes) and summarize immediate interpretation boundaries.

1 Execution scope

The following items were executed:

- Canonical QRNG pipeline run.
- Null-fixture adversarial run.
- Deterministic rerun and artifact comparison.
- Pipeline upgrades for statistical validity and preregistration readiness.

2 Repository and environment

- Repo: `toe-empirical-validation`
- Commit: `163418b9ff5410825133cffd75a5e0d37fbc1ce4`
- Contract status: current external campaign baseline; earlier `spine-v0.1` and `spine-v0.1.1` tags are legacy-only.
- Python: `3.14.1`
- Channel: `qrng`

3 Executed commands

```
./scripts/run_all.sh --channel qrng --seed 0 --out artifacts/
./scripts/run_all.sh --channel qrng --null-fixture --seed 0 --out artifacts_null/
./scripts/run_all.sh --channel qrng --seed 0 --out artifacts_repeat/
./scripts/compare_artifacts.sh artifacts/ artifacts_repeat/
```

4 Statistical pipeline upgrades

4.1 Primary test

The QRNG pipeline now computes:

$$T = \frac{1}{N} \sum_{b=1}^{N} \left(\overline{x}_b - \frac{1}{2} \right) E_b$$

where $\overline{x}_b$ is block mean and E_b = `label_E(bits)`. Significance is estimated via a two-sided permutation test (default `n_perm`=5000), with decision rule:

$$\texttt{effect_detected} = (\texttt{p_value} < \alpha), \quad \alpha = 0.05.$$

4.2 Operational E-label estimator

The canonical block estimator combines:

- bias magnitude $|\overline{x}_b - 0.5|$,
- run-length variability,
- edge-flip rate,

with fixed default weights $(0.5, 0.3, 0.2)$.

5 Observed outputs

5.1 Canonical run (`artifacts/constraints.json`)

Field	Value
`effect_detected`	`false`
`p_value`	0.3843231353729254
`alpha`	0.05
`test_statistic`	-0.00015285982327467018
`eta`	0.02
`n_perm`	5000
`n_blocks`	100
`block_size`	64
`seed`	0

5.2 Null fixture (`artifacts_null/constraints.json`)

Field	Value
`effect_detected`	`false`
`p_value`	1.0
`alpha`	0.05
`test_statistic`	0.0
`eta`	0.02
`n_perm`	5000
`n_blocks`	100
`block_size`	64
`seed`	0

5.3 Determinism check

The repeat run for seed 0 passed `compare_artifacts.sh` on `constraints.json` (within configured EPS) and required artifact presence checks.

6 Interpretation (current)

Current executed runs do not detect an effect at the preregistered threshold ($\alpha = 0.05$), and the null fixture correctly returns no effect. This is a pipeline-validity result, not a final empirical verdict. The framework remains in the stage where preregistered hardware acquisition and independent replication determine scientific closure.

7 Protocol artifacts added

- OSF-ready prereg template: `experiments/OSF_PREREG_QRNG_FLAGSHIP_V1.md`
- TeX build reproducibility notes: `TOE/papers_sources/BUILD_REPRODUCIBLE_2026.md`

8 Next execution gates

1. Run preregistered hardware QRNG acquisition under fixed estimator and decision rule.
2. Publish artifact bundle (raw data, checksums, commands, commit hash, outputs).
3. Execute independent replication at one external lab/group.
4. Integrate result into canonical falsification packet and manuscript update.